Hongwei CUI

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willing.website
Hong Kong, China

EDUCATION

- **Ph.D. Information Engineering** *2021 - Present*
  The Chinese University of Hong Kong (QS #32)
  Supervised by Prof. Soung Chang Liew

  Interested Research Area:

  Optical Wireless Communication
  AI-powered Wireless Sensing
  LLM-based AI Agents
  Embedded Systems and IoT Systems

- **B.S. Electronic Information Engineering** *2017 - 2021*
  Shandong University
  CGPA: 4.625/5.0 (Ranking 1st in Chongxin College)

INTERNSHIP

- **Researcher** *2025/10 - Present*
  Huawei Noah's Ark Lab (Hong Kong Research Center)
  Embodied Intelligence & Wireless Sensing

AWARDS & RECOGNITION

- **First Prize of Taishan Student Scholarship**
  2024
  Tai Shan Students Foundation

SKILLS

- **Research and Development Abilities**
  - Transformer architecture expertise: end-to-end AI model development, optimization, deployment.
  - Web front-end & mobile app development.
  - Full-stack IoT & embedded systems development.
  - High-speed analog/digital circuit design, simulation, PCB layout, soldering, and debugging.
  - 3D modeling & product rendering.

- **Programming Languages/Frameworks**
  C, C++, Python, Pytorch, HTML5, JavaScript,
  React Native, Verilog, Java

RESEARCH WORKS

- **Optical Wireless Ether: Enabling Controlled Dynamic Signal Propagation in OWC Systems**
  Hongwei Cui, Soung Chang Liew
  arXiv (Preprint)
  2025/2/10

- **Wi-LiFi: Integrated Optical Wi-Fi for Enhanced Mobile Robotic Communications and Localization**
  Hongwei Cui, Soung Chang Liew, He Chen
  IEEE Transactions on Vehicular Technology
  2024/12/13

- **LLMind: Orchestrating AI and IoT with LLM for Complex Task Execution**
  Hongwei Cui, Yuyang Du, Qun Yang, Yulin Shao, Soung Chang Liew
  IEEE Communications Magazine
  2024/9/27

- **Wi-LiFi: Integrated Optical Wi-Fi for Enhanced Mobile Robotic Communications**
  Hongwei Cui, Soung Chang Liew, He Chen
  IEEE International Conference on Communications
  2024/6/9

PATENT

- 一种**Wi-Fi**信道检测系统及方法 **(A Wi-Fi Channel Detection System and Method)**
  崔洪玮 (Hongwei Cui)
  Patent Public Notice No.: CN114448529A
  2022/5/6

HOBBIES & INTERESTS

Electronic Circuit | AI & LLM | UI & UX | Photography